

INTEGRATED PETROCHEMICAL COMPLEX & INFRASTRUCTURE PROJECT, UNIT PDH KAZAKHSTAN

PRELIMINARY GEOTECHNICAL AND FOUNDATION STUDY REPORT

Fugro Report No. C120445/01

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Foster Wheeler Engineering & Construction Design (Shanghai) Co. Ltd



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1 INTRODUCTION

Fugro Comprehensive Geotechnical Investigation (Zhejiang) CO., Ltd. ("FZJ") was contracted by Foster Wheeler Engineering & Construction Design (Shanghai) Co. Ltd ("FW") to carry out a geotechnical and foundation study for the Area PDH of the Integrated Petrochemical Complex and Infrastructure (IPCI) Project. This project is a joint project of Sinopec with the Kazakhstan Petro-chemical Industry. The scope of works was initiated by FW in the Contract No. 1CD1305A-SO-01. The scope of works is also listed in Section 1.2.

The study would be based on information provided by FW and is subject to the accuracy of the information. This is a preliminary report which covered the site conditions, geotechnical conditions and the pile capacities as requested by FW. The information will be combined with the other results of the study in the final report.

1.1 Project Description

The IPCI Project area is situated in Makat region of Atyrau Oblast, about 47km from Atyrau city, Republic of Kazakhstan.

The Area PDH is about 100,000m² (370m x 270m). The proposed facilities include buildings and various petro-chemical structures. **Plate 1** presents the location of Atyrau. **Plate 2** presents the general equipment layout.

1.2 Scope of Works

As given in the Contract No. 1CD1305A-SO-01, Scope of Works includes the followings:

- (1) Existing report analysis
 - (a) Comment on the site investigation results;
 - (b) Comment on the geology of the sections. If the locations of the boreholes, Static Penetration Tests and sections are known then some of the results will be added to the geological sections;
 - (c) Collect geological information of the region and comment on the site investigation results;
- (2) Preliminary foundation assessment
 - (a) Propose the additional geotechnical and geophysical investigations for preliminary foundation assessment and for detailed foundation designs;
 - (b) If necessary arrange and supervise the additional investigations;

- (c) The suitability to use different types of deep and shallow foundations;
- (d) The bearing capacity for the preliminary designs of deep and shallow foundations;
- (e) Potential settlement of deep and shallow foundations;
- (f) Availability of construction plan in the region;

(3) Pile capacity estimation

Make preliminary estimation of the size and depth of precast and bored pile to achieve the following:

- (a) Pile ultimate capacity of 50 tonnes;
- (b) Pile ultimate capacity of 100 tonnes;
- (c) Pile ultimate capacity of 150 tonnes;
- (d) Pile ultimate capacity of 300 tonnes;

(4) Soil liquefaction analysis; and

(5) Settlement analysis

To select 3 typical soils profiles and assume 2 sizes of shallow foundations for the settlement analysis

Based on Fugro's proposal dated 13 June 2012, this preliminary report presents the contents on pile capacity estimation. A detailed report will be submitted to FW in due course on other assignments.

1.3 Available Information

The information supplied by FW for this study is listed in the following Table 1:

Table 1: List of Information Provided by FW

Item	Description	Reference No.
1	Caspian Institute of Exploration Geophysics, Engineering-geological survey materials, Integrated Petrochemical Complex, Report. Attachment	Ref 1
2	Tetraconengineering, Engineering-geological surveys, Topographic plan, Engineering-geological cuts, Static penetration test records	Ref 2
3	Tetraconengineering, Final Report attachment D, E, F	Ref 3
4	Sinopec Shanghai Engineering Company Ltd, General Equipment Layout	Ref 4

The current scope for this review and study is limited to the above design documents provided.

1.4 Design Standards

Fugro strives to perform the above mentioned services in accordance with recognised international standards such as API, ASTM and BS as referred to in reference.

2 SITE CONDITIONS

2.1 Available geotechnical results

The available investigation results were carried by 2 Kazakhstan companies in 2009 (Ref 1) and 2012 (Ref 2, Ref 3), respectively. The report in 2009 (Ref 1) is a comprehensive geotechnical report for another site of the project. The results in Ref 2 and Ref 3 are only factual records of a geotechnical investigation for the site covered by this study. The report was prepared in accordance with Russian standards (SNIP and GOST). These Russian standards are close to Chinese standards. Basic principles of those standards in the geotechnical field are comparable to standards such as ASTM and BS but differences exist. For this particular project the Russian style report has been re-worked to a ASTM style result.

The fieldworks of the investigation in Ref 1 included 27 boreholes and 27 CPT test points. However, the exact locations of these tests are not provided. Thus, most of the geotechnical parameters in this report are used for references only.

The available fieldworks of the investigation in Ref 2 included 18 CPT tests as listed in Table 2. The borehole and CPT locations are presented in **Plate 3**. In general, the investigation program included sufficient coverage of the site. However, the available investigated depth is only 30m below ground surface. Although it was shown that some boreholes were sunk to 45 to 50m below ground, these results were not provided, leaving significant parts of the sub soil parameters unquantified.

Table 2: Available CPT Tests

CPT No.	Depth (m)	CPT No.	Depth (m)	CPT No.	Depth (m)
1211	30	1241A	30	1262	30
1215	30	1243	30	1266	30
1221	30	1245	30	1268	30
1225	30	1247	30	1269	30
1238	30	1257	30	1272	30
1239	30	1259	30	1273	30

Fugro observed some cone tip resistance q_c values seem doubtful. An example is CPT1262 showing low q_c even zero values. Such readings might be indicative of some problem with the recording, or, since the original ASCII data was not provided, may represent rounding down to zero of low q_c readings.

2.2 Site Setting

2.2.1 Surface Conditions

Generally, the terrain at the proposed site is flat and is part of the Caspian coastal semi desert. Elevations of the existing relief relative to the North Baltic Sea datum vary from minus 20.53m to minus 22.78m (Ref 2).

2.2.2 Climate

The climate of the region is clear continental and arid. A high temperature contrast exists between days and nights in both summer and winter. There is a rapid change from winter to summer as spring is very short. The weather is changeable with little precipitation and snow, with typically dry air and soils exposed to abundant sunlight and evaporation. The annual amount of precipitation in winter time is approximately 70mm, and slightly higher 102mm in the summer time. Winter is cold and long with the minimum temperature of -38°C. Summers are hot and rather long with the maximum temperature reaching +45°C. Local practice considers frost action limited to -1.24m for fine-grained soils and -1.5m for fine and silty sands. The effect of Caspian Sea presence on the local climate conditions over the study site is negligible (Ref 1).

2.3 Laboratory Testing

The available geotechnical laboratory test results in Ref 3 comprise:

- Classification testing: water content, unit weight, particle size distribution, Atterberg limits, density of solid particles,
- Compressibility testing: incremental loading in oedometer,
- Strength testing: direct shear tests and triaxial tests,

The above listed tests have been performed in accordance with SNiP and GOST standards. The interpretation and use of such test results require caution in BS/ASTM style engineering assessments, due to some differences in methods and definitions. Interpretation of laboratory test results requires an assessment of sample quality. Sample quality classes relate to feasibility and interpretation of specific geotechnical laboratory tests. It is not possible to assess sample quality from the provided geotechnical report (Ref 3).

For the engineering study presented in this report, Fugro assumes that samples quality is generally adequate for most geotechnical classification tests.

A summary of the test results is presented in **Plate 4**.

Nine unconsolidated undrained tests are presented in Ref 3 and listed in Table 3. The location of Borehole No. 4024 is not provided. Other tests are for samples 30m below ground surface. Reliance has had to be placed on the CPT test data to determine the undrained shear strength and relate that to the previous stress history at the site. Ideally additional testing would have been carried out to calibrate and verify the interpretation of the CPT data. It is recommended that such testing should be completed in the future phase of the site investigation, comprising a series of conventional consolidated/unconsolidated undrained triaxial tests with pore pressure measurement to determine both undrained shear strength and effective angle of friction for the clay units.

Table 3: Unconsolidated Undrained Tests

Borehole No.	Depth (m)	Soil Type	EGE No	Density (kN/m ³)	C _u (kPa)
4042	20.6	Sandy Clay	6	20.4	123
9005/1	32	Very stiff Clay	9	19.7	147
1262/2	32	Stiff Clay	9	17.8	95
2002/3	33	Clay with silt	9	19	7
1256/2	34	Stiff Clay	9	17.6	85
1271/4	38	Hard Clay	9	20.7	608
1271/6	43.5	Hard Clay	9	19.4	224
9010/6	44	Sandy Silt	9	18.7	472
1262/8	45	Very stiff Clay	9	20.5	197

2.4 Stratigraphic Units

Engineering-geological elements (EGE) classified in Ref 2 are retained to be consistent with the GI report. These EGEs are described below.

EGE 1. Sandy Clay

EGE 2. Brown Clay with inclusion of carbonate

EGE 3. Gray Clay with layers of sand

EGE 4. Gray sandy Clay with inclusion of carbonate shells

EGE 5a. Dense brownish gray Sand

EGE 6. Bluish-gray clay with layers of sand

EGE 7. Dark gray sandy Clay

EGE 8. Dense gray Sand

EGE 9. Dark gray Clay with layers of sand

EGE 10. Dense gray Sand with inclusion of seashells

Laboratory test data presented in Ref 3 do not include the method statements for the tests. The assumption was made that all tests have been performed in accordance with SNIIP/GOST standards.

Soil classification is generally based on grain size and Atterberg limits. In GOST and ASTM/BS, the definitions of silt and silty clay determined from the particle size distribution curves are different.

According to GOST, silt and silty clay are defined by the plasticity index $I_p > 1$. Further classification is based on plasticity index, contents of 0.05 - 2-mm fractions, and liquidity index I_L .

According to ASTM/BS, cohesive soils are determined by the percent content of fine particles with diameters of less than 0.075mm (ASTM) or 0.063mm (BS). Further classification is based on plasticity chart.

According to Chinese standards, cohesive soils are determined by the percent content of fine particles with diameters of less than 0.075mm. Further classification is based on plasticity index.

The SNIIP/GOST uses the Vasilev test method to determine the liquid limit (w_L , %) while the ASTM/BS uses the Casagrande method. Liquid limit tests by Vasilev show a bit lower results compared to the Casagrande method. The liquid limit adopted in Chinese standard is close to the liquid limit with the test method developed by Vasilev. The Atterberg limits in Ref 3 are plotted in the plasticity chart in **Plate 5**. By examining these results and the grain size distribution, EGE 1 and EGE 4 are very likely to be sandy SILT.

2.5 Groundwater Conditions

According to Ref. 2, groundwater levels are at 3.4 to 4.6m below the ground surface. The soil and ground water are highly aggressive to concrete (Ref 1).

3 GEOTECHNICAL ASSESSMENT

3.1 Geological Sections

The engineering-geological elements classified in Ref 2 are used for this study. The CPT results for Section 3-3, Section 5-5 and Section 9-9 which are representative to the general conditions are presented in **Plates 6 to 8**. The stratigraphic units for Section 5-5 are shown on **Plate 9**.

3.2 Relative Density

To assess the density conditions of cohesionless soils a detailed inspection was performed on the results of cone penetration tests. Local peaks observed in CPT cone resistance and CPT refusal probably relate to (thin) layers of cemented soil or gypsum.

Interpretation of the relative density, D_r is based on CPT data by using equation by Jamiolkowski et al. (1988):

$$D_r = \frac{1}{2.93} \ln \left[\frac{q_c}{205(p')^{0.51}} \right]$$

where

q_c is the cone tip resistance in kPa,

$$p' = (\sigma'_{v0} + 2\sigma'_{h0}) / 3 \text{ [kPa]}$$

σ'_{v0} = effective in-situ vertical stress in kPa,

σ'_{h0} = the effective in-situ horizontal stress in kPa.

3.3 Undrained Shear Strength from CPT q_c Data

The conventional analysis for undrained shear strength uses the following relation:

$$S_u = (q_c - \sigma_{v0}) / N_k$$

where

S_u = the undrained shear strength,

σ_{v0} = the current vertical (total) stress, and

N_k = cone factor.

For the present interpretation, a cone factor $N_k = 20$ has been used to derive representative undrained shear strength parameters for design.

3.4 Parameters for Analysis

The records of CPT1247 are used as a representative test results for this analysis. The derived strength parameters are presented on **Plate 10**. It is noted that it may be better to refine classification of the upper soil layers.

4 REPRESENTATIVE PILE CAPACITIES CURVES

4.1 General

This section presents considerations, analyses and recommendations for driven full displacement piles and bored cast in-situ piles. Driven pile analysis follows the procedure of API (2000, 2011). Bored pile analysis follows the methods in Canadian Foundation Engineering Manual (1992) and Bowles (1996). Bottom of pile cap is fixed at 2m below ground level to account for the frost depth and soil contamination. The derived parameters at CPT1247 are used for the analysis.

4.2 Axial Capacity of Single Piles in Soil

The ultimate axial capacity of single piles in compression is determined by the following equation:

$$Q_u = u \sum q_{si} l_i + q_b A_p$$

where:

Q_u	Ultimate single pile bearing capacity,
q_{si}	Skin friction capacity,
q_b	End bearing capacity,
A_p	Pile sectional area,
u	Pile perimeter,
l_i	Thickness of the i^{th} soil layer,

4.3 Axial Capacity for Driven Piles

4.3.1 Shaft Friction and End Bearing in Cohesive Soils

For piles in cohesive soils, the shaft friction at any point along the pile is calculated by the equation.

$$q_s = \alpha S_u$$

where:

$$\begin{aligned} \alpha &= 0.5\psi^{-0.5} & \text{for } \psi \leq 1 \\ \alpha &= 0.5\psi^{-0.25} & \text{for } \psi > 1 \\ \psi &= S_u / \sigma'_{v0} \end{aligned}$$

σ'_{v0} Effective in-situ vertical stress at the point in question,

For piles end bearing in cohesive soils, the unit end bearing is computed by the equation.

$$q_b = 9 S_u$$

4.3.2 Shaft Friction and End Bearing in Cohesionless Soils

For piles in cohesionless soils, the shaft friction at any point along the pile is calculated by the equation.

$$q_s = K \sigma'_{v0} \tan \delta$$

where:

K Coefficient of lateral earth pressure,
 σ'_{v0} Effective in-situ vertical stress at the point in question,
 δ Friction angle between the soil and pile.

For piles end bearing in cohesionless soils, the unit end bearing is computed by the equation.

$$q_b = N_t \sigma'_{v0}$$

where:

N_t Bearing capacity factor.

Recommended values for driven piles can be found in API (2000, 2011).

4.3.3 Representative Pile Capacity Curves of Driven Piles

Fugro parameter values in this calculation are shown on **Plate 11**. The values for ultimate unit end-bearing and skin friction versus depth computed for the driven piles are shown on **Plate 12**. The results for compressive axial loading conditions for 300mm-diameter, 400mm-diameter and 500mm-diameter driven precast concrete pipe piles are presented on **Plates 13 to 15**. The presented ultimate axial pile capacity results do not account for the effective weight of the pile in the soil (i.e. pile weight must be considered as a load) and do not include any safety factors.

4.4 Axial Capacity for Bored Piles

4.4.1 Shaft Friction and End Bearing in Cohesive Soils

For piles in cohesive soils, the shaft friction at any point along the pile is calculated by the equation as suggested in Canadian Foundation Engineering Manual (1992).

$$q_s = \beta \sigma'_{v0}$$

where:

β Dimensionless shaft friction factor,

For piles end bearing in cohesive soils, the unit end bearing is computed by the equation.

$$q_b = 9 S_u$$

4.4.2 Shaft Friction and End Bearing in Cohesionless Soils

For piles in cohesionless soils, the shaft friction at any point along the pile is calculated by the equation.

$$q_s = \beta \sigma'_{v0}$$

For piles end bearing in cohesionless soils, the unit end bearing is computed by the equation.

$$q_b = N_t \sigma'_{v0}$$

4.4.3 Representative Pile Capacity Curves of Bored Piles

Fugro parameter values in this calculation are shown on **Plate 16**. The results for compressive axial loading conditions for 600mm-diameter and 800mm-diameter bored piles are presented on **Plates 17 and 18**. The presented ultimate axial pile capacity results do not account for the effective weight of the pile in the soil (i.e. pile weight must be considered as a load) and do not include any safety factors.

5 SUMMARY AND RECOMMENDATIONS

Based on the existing information and analysis above, Fugro's recommendations are suggested as follows:

- (1) Driven precast concrete pipe piles with a diameter of 300mm sunk to about 15m below ground surface are recommended for pile capacities of 50 and 100 Tonnes;
- (2) Driven piles with a diameter of 400mm sunk to about 15m below ground surface are recommended for pile capacities of 150 Tonnes;
- (3) Driven piles with a diameter of 400mm sunk to about 28m below ground surface are recommended for pile capacities of 300 Tonnes;
- (4) Bored piles with a diameter of 800mm tipped at about 15m below ground surface are recommended for pile capacities of 50, 100 and 150 Tonnes;
- (5) Bored piles with a diameter of 800mm tipped at about 20m below ground surface are recommended for pile capacities of 300 Tonnes;
- (6) Several deep boreholes for quantifying the underlying stratum below 30m should be provided or arranged;
- (7) The existing layer classification may need to be refined;
- (8) Some laboratory test results such as unconsolidated undrained strength for the upper clay layers should be provided or arranged;

6 APPLICABILITY OF REPORT

The assessment and analysis performed for this study, as well as the conclusions and recommendations in this report, were developed based on the review of the available documents and information as described in this report and provided by the Client. Fugro does not accept any responsibility for the accuracy of the data contained within the documents and information made available by the Client.

The observations, conclusions, and recommendations presented in this report may not apply to situations that differ from those presented in the report. If pertinent details of the project have changed or otherwise differ from those provided to us, Fugro requests that it be engaged to review such changes and to modify our recommendations, if found necessary.

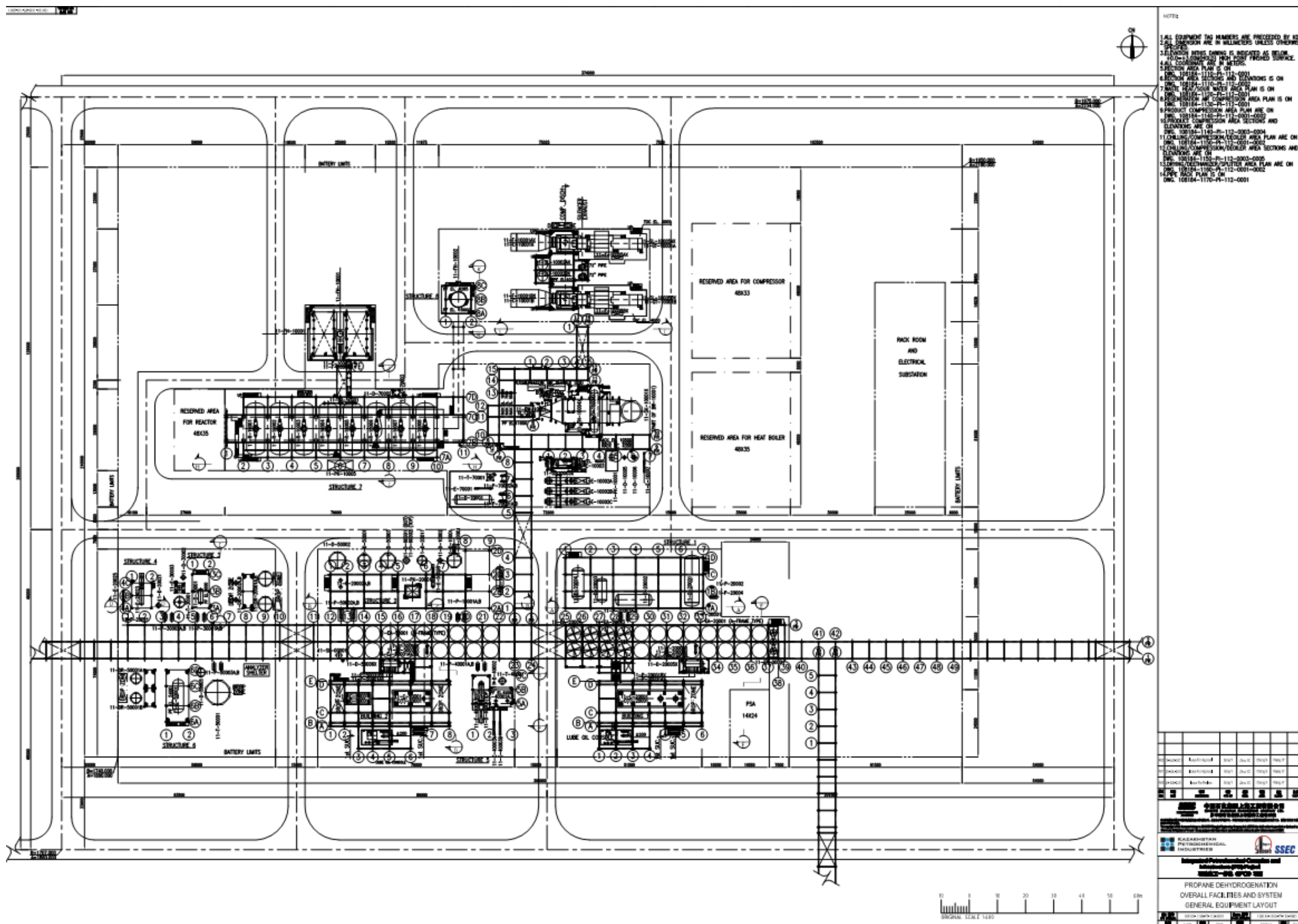
Fugro has prepared this report exclusively for Foster Wheeler Engineering & Construction Design (Shanghai) Co. Ltd. This report is intended to be used for the purposes describes in Introduction. The report is not intended to be used for any other purpose. Fugro makes no claim or representation concerning any activity or conditions falling outside the specified purpose to which this report is directed.

7 REFERENCE

- (1) Caspian Institute of Exploration Geophysics, Engineering-geological survey materials, Integrated Petrochemical Complex, Report. Attachment, REV D3, (File No. KIRG_ATL-36-2006_48533_IPC_GEOTEC_REPORT_RevD3.pdf) dated 31/03/2009
- (2) Tetraconengineering, Engineering-geological surveys, Topographic plan, Engineering-geological cuts, Static penetration test records, (File No. PDH Unit 11.zip) dated Feb 2012
- (3) Tetraconengineering, Final Report attachment D, E, F (File No. Final Report_Att D-E-F.pdf) dated Feb 2012
- (4) Sinopec Shanghai Engineering Company Ltd, General Equipment Layout, R02, dated 19/08/2011
- (5) Jamiolkowski, M., Ghionna, V.N., Lancellotta, R. and Pasqualini, E. (1988), "New Correlations of Penetration Tests for Design Practice", in De Ruiter, J. (Ed.), Penetration Testing 1988: Proceedings of the First International Symposium on Penetration Testing, ISOPT-1, Orlando, 20-24 March 1988, Vol. 1, A.A. Balkema, Rotterdam, pp. 263-296.
- (6) American Petroleum Institute (2011). Geotechnical and Foundation Design Considerations, API Recommended Practice 2GEO, 1st Edition
- (7) American Petroleum Institute (2000). Recommended Practice for Planning, Designing and Construction Fixed Offshore Platforms, API Recommended Practice 2A-WSD, 21st Edition.
- (8) Canadian Geotechnical Society (1992). Canadian Foundation Engineering Manual, 3rd Edition.
- (9) Bowles, J. E. (1996). Foundation Analysis and Design, 5th Edition
- (10) GB50021-2001 (2009). Code for Investigation of Geotechnical Engineering.
- (11) ASTM D2487-10 (2010). Standard Practice for Classification of Soils for Engineering Purposes.



VICINITY MAP
IPCI Project - Kazakhstan



Notes:

- Reproduced from Client
- Not to scale

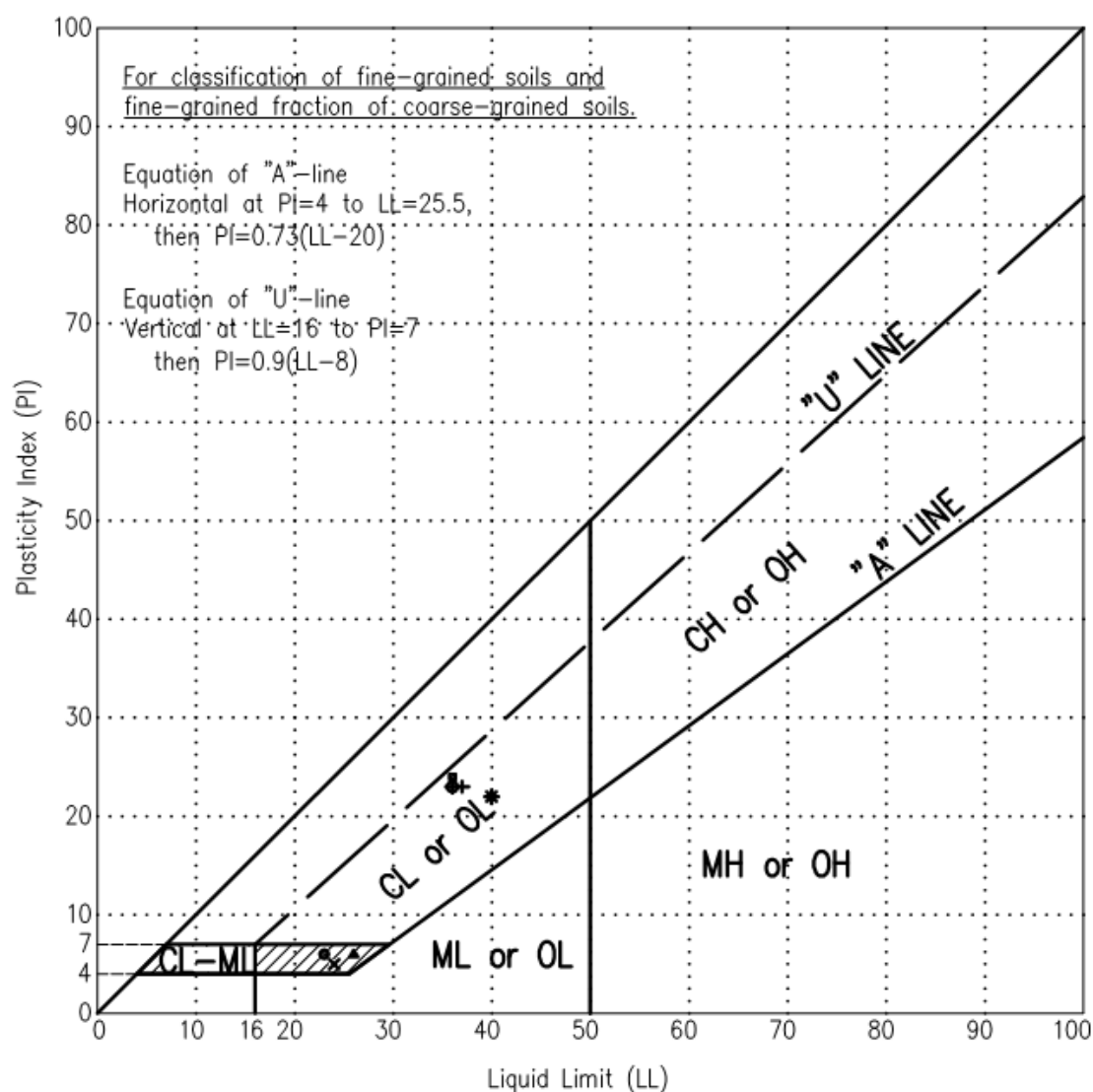
PDH EQUIPMENT ARRANGMENT LAYOUT PROVIDED BY CLIENT
IPCI Project - Kazakhstan

Summary of Laboratory test results - standard values (1)

EGE No	Soil name (from KPI report)	Granulometry structure (%)									Soil density (g/cm ³)			Submerged unit weight (kN/m ³)	Void ratio	Water content, W		
		Gravel		Sand				Dust		Clay								
		>10	10-2	2-0.5	0.5-0.25	0.25-0.1	0.1-0.05	0.05-0.01	0.01-0.005	<0.005	Natural	Dry	Parts			Natural	Liquid limit	Plastic limit
											ρ	ρ_d	ρ_s					
											ρ	ρ_d	ρ_s	γ_{sb}	e	W	W_L	W_P
1	Sandy loam solid										1.77	1.60	2.70	10.09	0.690	0.13	0.24	0.19
1A	Pulverescent sand		1.8	7.5	24.3	34.0	32.2				1.60	1.44	2.65	8.94	0.850	0.10		
2	Clay semi-solid										1.88	1.58	2.74	10.06	0.730	0.19	0.37	0.14
3	Clay flow-plastic										1.85	1.52	2.74	9.61	0.810	0.22	0.36	0.13
4	Sandy loam plastic										1.86	1.53	2.68	9.58	0.750	0.21	0.23	0.18
5	Pulverescent sand		2.0	7.8	24.1	31.1	35.0				1.80	1.47	2.65	9.15	0.800	0.25		
5a	Pulverescent sand		0.3	4.8	23.1	35.8	36.1				1.90	1.51	2.66	9.41	0.760	0.23		
6	Clay flow-plastic										1.85	1.52	2.74	9.65	0.800	0.22	0.36	0.12
7	Sandy loam plastic										1.84	1.48	2.70	9.29	0.830	0.25	0.26	0.20
8	Pulverescent sand		1.1	5.8	25.5	34.6	33.0				1.90	1.51	2.67	9.43	0.770	0.23		
9	Clay loam solid										1.86	1.51	2.74	9.59	0.810	0.23	0.40	0.18
10	Pulverescent sand		2.1	9.4	23.3	33.4	31.9				1.84	1.51	2.65	9.39	0.758	0.22		

Summary of Laboratory test results - standard values (2)

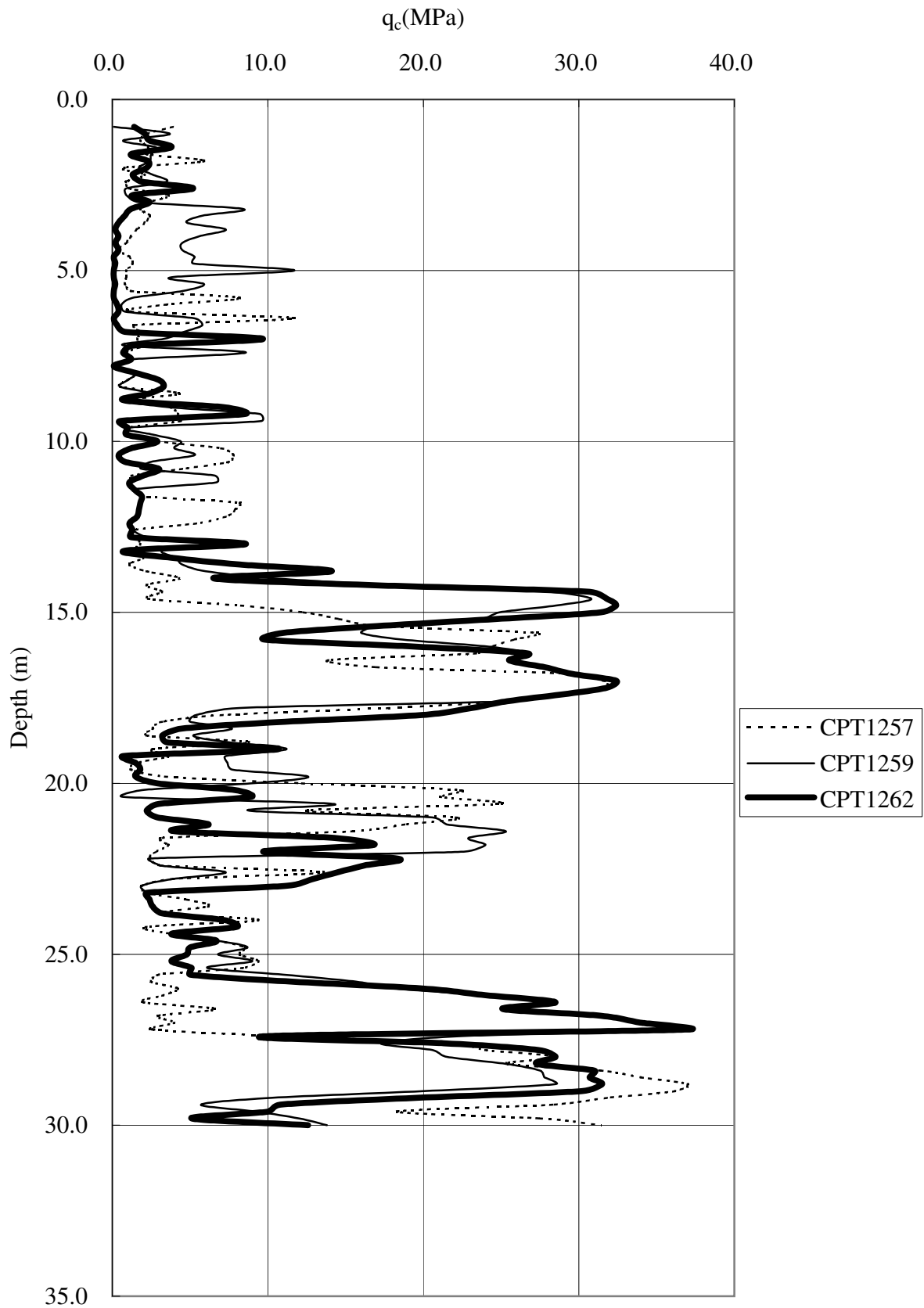
EGE No	Soil name	Plastic Index	Liquidity index	Sand density (g/cm ³)		Sand front-to-back slope (degree)		Degree of saturation	Filtration coef. (m/day)	Relative cont. of organic subst. (u. s.)	Shear σ = 0-0.3 (MPa)				Deformation module E(Mpa), the pressure, p=0.1-0.2 (MPa)				Subsidence of soil (dilatation) at full water-saturation without counting $K_{s1,i}$ (sw, i)	
											Internal friction angle (dg.)		Cohesion(kPa)		Compression		General			
											Natural	Saturate	Natural	Saturate	Natural	Saturated	Natural	Saturated		
		I_P	I_L	ρ_{max}	ρ_{min}	Dry	under water	S_r	k_Φ	I_{om}	φ		c						$\varepsilon_{sl,i} (sw, i)$	Δh
1	Sandy loam solid	0.05	-0.86					0.490			24.769		6.833		4.23		13.10			
1A	Pulverescent sand			1.70	1.41	33.60	25.70	0.310			26.470		2.570		9.84					
2	Clay semi-solid	0.23	0.24					0.690			17.084		35.704		3.13		18.40			
3	Clay flow-plastic	0.23	0.39					0.750			16.930		32.773		3.30		18.28			
4	Sandy loam plastic	0.06	0.69					0.760			25.965		5.296		4.23		10.90			
5	Pulverescent sand			1.87	1.43	36.20	26.80	0.820			26.120		2.920		9.74					
5a	Pulverescent sand			1.90	1.57	37.17	28.33	0.810			25.170		3.380		20.52					
6	Clay flow-plastic	0.24	0.42					0.760			15.169		33.633		3.66		20.53			
7	Sandy loam plastic	0.06	0.81					0.800			21.148		3.929		3.01		6.45			
8	Pulverescent sand			1.92	1.55	37.17	27.00	0.810			27.100		2.660		24.46					
9	Clay loam solid	0.22	0.21					0.770			19.322		36.861		4.46		24.69			
10	Pulverescent sand			1.90	1.43	37.13	27.50	0.768			24.910		2.310		13.11					



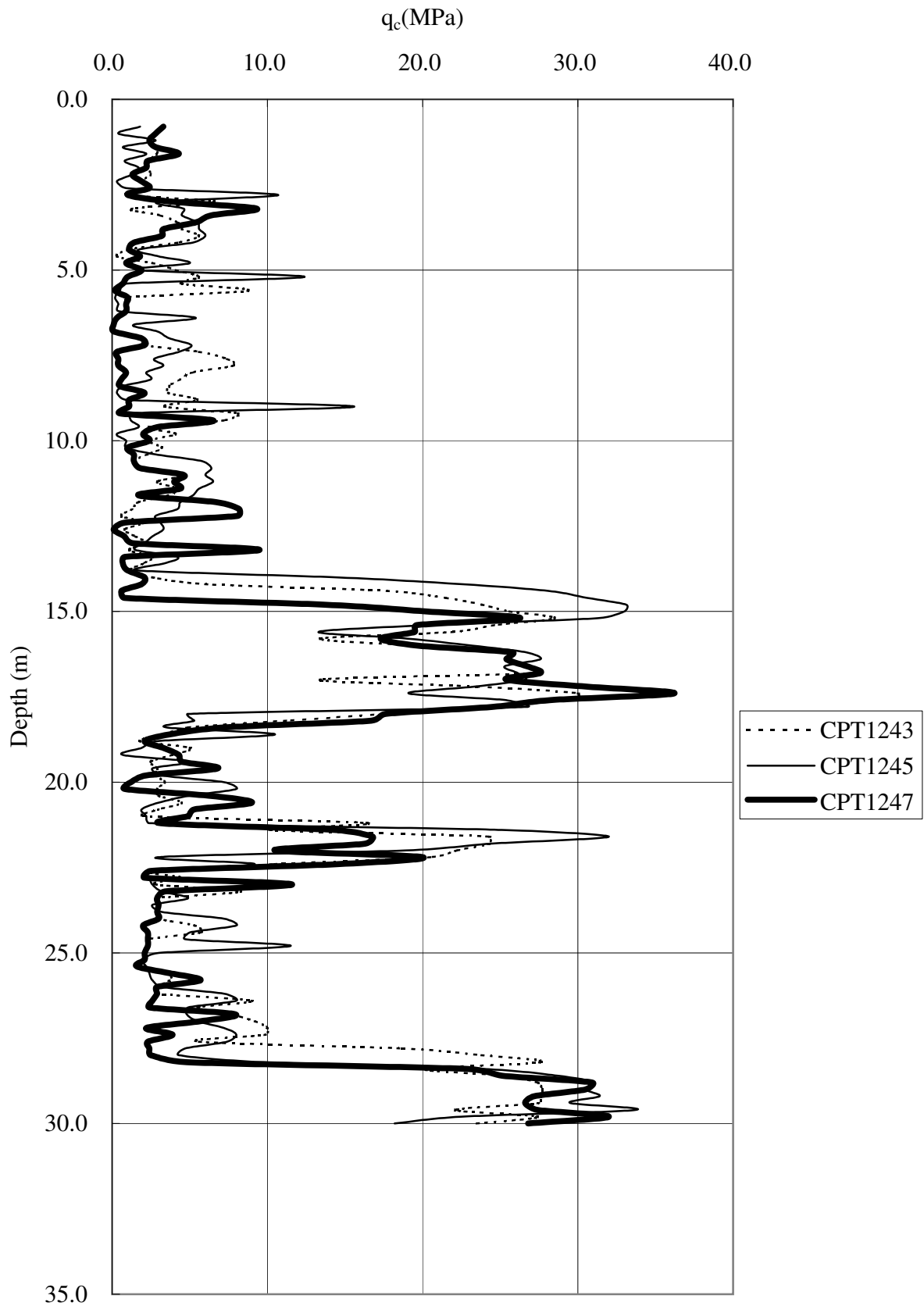
LEGEND :

- | | | | |
|------------|------------|------------|------------|
| × EGE NO.1 | ◆ EGE NO.3 | ■ EGE NO.6 | * EGE NO.9 |
| + EGE NO.2 | ● EGE NO.4 | ▲ EGE NO.7 | |

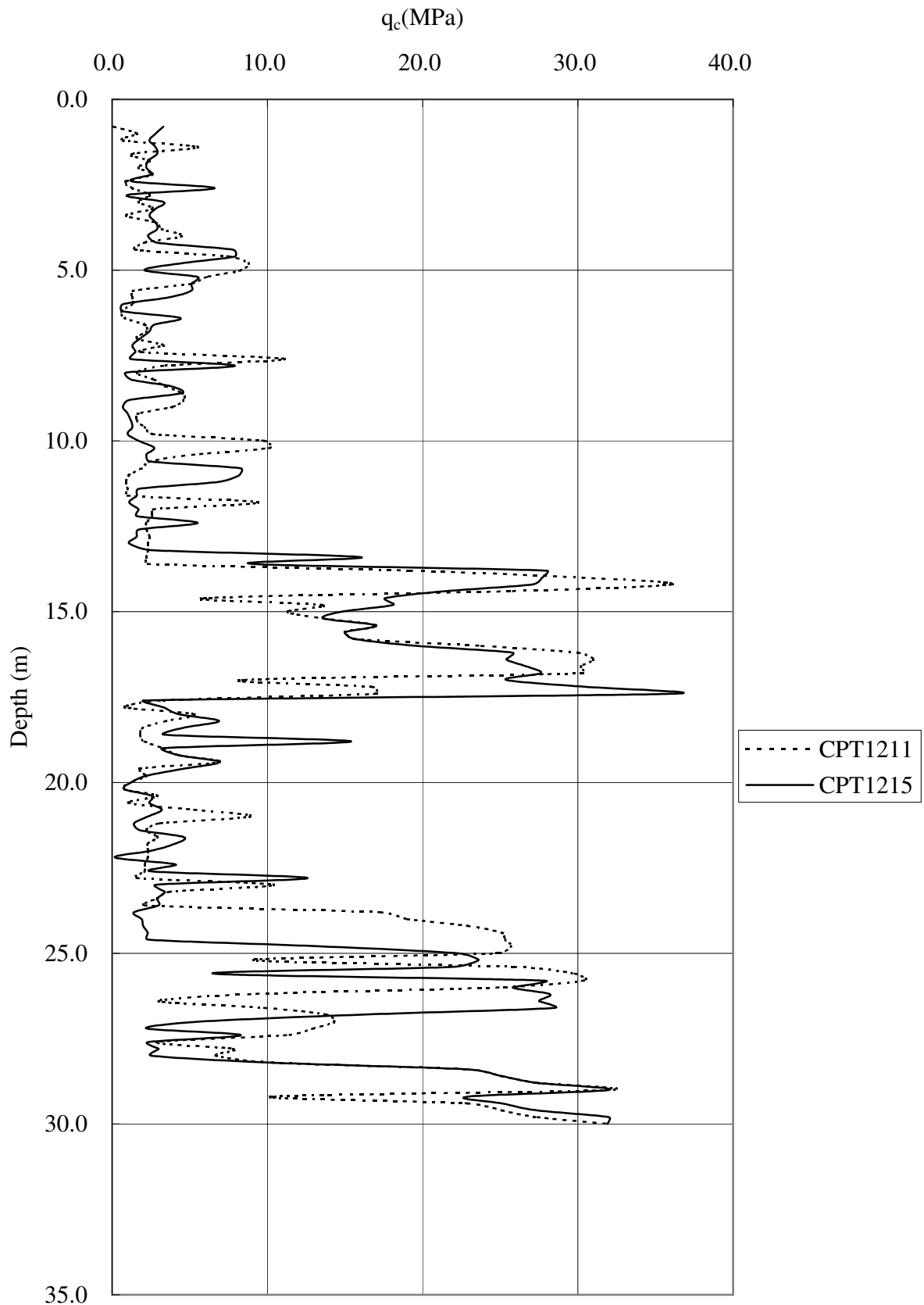
PLASTICITY CHART
IPCI Project - Kazakhstan



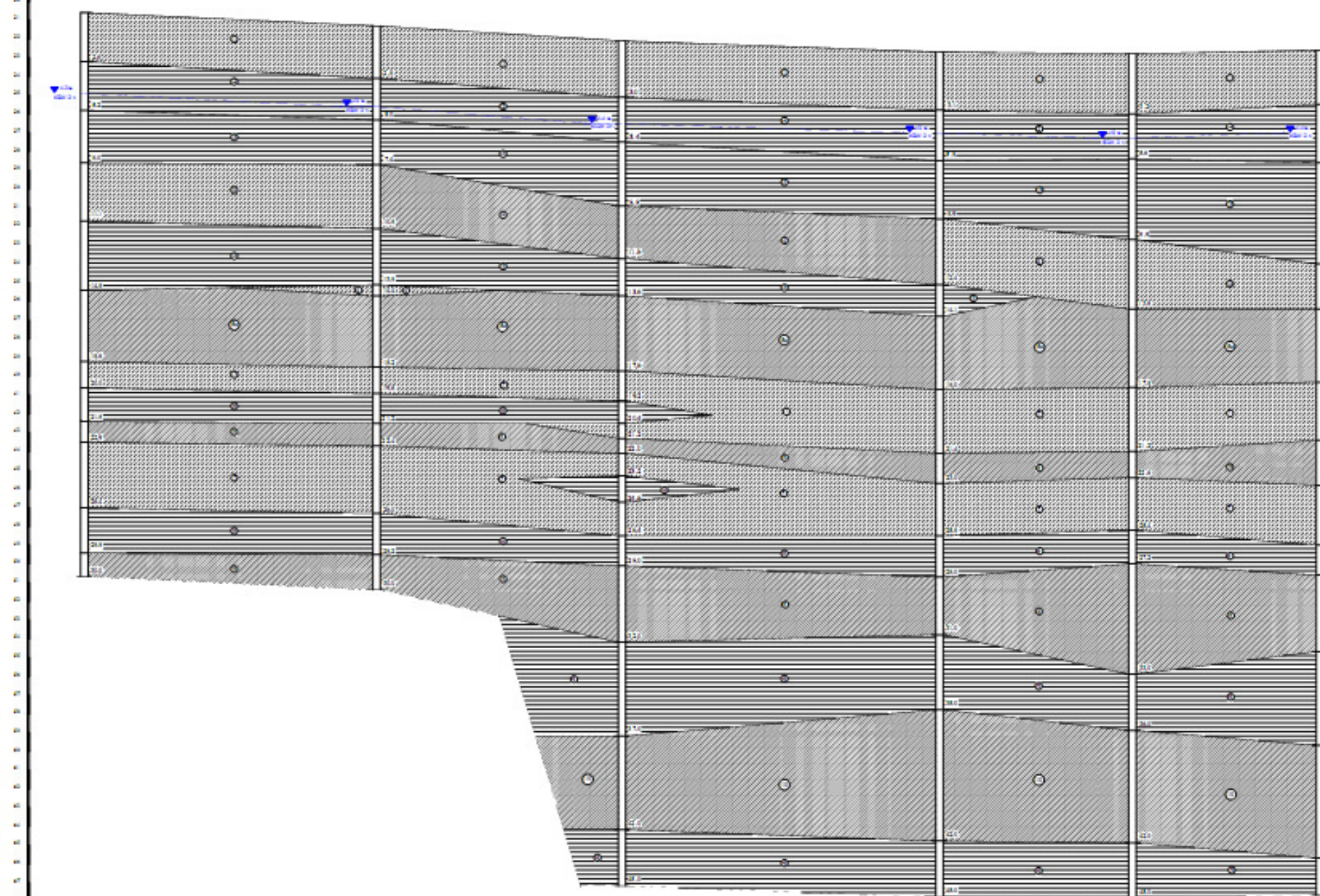
CPT CONE RESISTANCE q_c FOR SECTION 3-3



CPT CONE RESISTANCE q_c FOR SECTION 5-5



CPT CONE RESISTANCE q_c FOR SECTION 9-9



Условные обозначения / LEGENDS:	
Q ₃ Hv	 Brown sandy loam, brown yellowish-gray with siliceous sandlayers of sand, clay loam with the inclusion of carbonates. Сухая, пористая, желтовато-серая с тонкими прослойками песка, глинистыми включением карбонатов.
	 Brown silty sand, silty with siliceous nodules, shells, clay, medium density. Буровато-серый, пылеватый с включениями карбонатов, ракушек, глинистый, средней плотности.
	 Brown clay with thin inclusion of carbonates. Глина бурого включения карбонатов.
	 Gray clay with sand and silt, plastic "mass". Глина серая с прослойками песка, пылеватой глины.
Q ₂ H2	 Gray sandy loam with the inclusion of carbonate shells, sandbed with layers of silt. Серая супесь с включениями раковин, пылеватая, разношерстная с прослойками глины.
	 Sand is brownish gray, pulverulent with thin inclusion of carbonates, shells, clay, dense. Песок буровато-серый, пылеватый с включениями карбонатов, ракушек, глинистый, плотный.
	 Brownish gray sand, pulverulent with carbonates inclusions, shells, clayey, dense. Песок буровато-серый, пылеватый с включениями карбонатов, ракушек, глинистый, плотный.
	 Mottled gray clay with layers of sand, the inclusion of gypsum, carbonates, limonized. Глина пестро-серая с прослойками песка, включениями гипса, карбонатов, окисленная.
N ak-up	 Dark gray sandy loam with the inclusion of shells, sanding. Сухая темно-серая с включениями ракушек, разношерстная.
	 Gray sand, fine, pulverulent, dense. Песок серый, тонкий, пылеватый, плотный.
	 Clay gray, dark gray to black with layers of sand, sandy loam, sand. Глина серая, темно-серая до черной с прослойками песка, пылеватая, песчаная.
	 Gray, fine, pulverulent and with the inclusion of seashells, dense. Песок серый, тонкий, пылеватый с включениями ракушек, плотный.

Consistency of soils Кonsistence gruyon				Degree of fluidity Степень текучести
Sandy loam Сыпучий Clay loam Глинистый Clay Глина	Cyrnec Сыпучий Глинистый Глина	Solid / Tvernyye	—	—
Sand Песок	—	—	—	Small degree of water retention Малое удержание влаги
Clay loam Сыпучий Clay Глина	Cyrnec Сыпучий Глинистый Глина	Semi-solid / Polutvernyye	—	—
Clay loam Сыпучий Clay Глина	Cyrnec Сыпучий Глинистый Глина	Low-plasticity Тугопластичность	—	—
Sandy clay Сыпучий Clay Глина	Cyrnec Сыпучий Глинистый Глина	Plastic Пластичность	—	—
Sand Песок	—	—	—	Very small degree of water retention Очень малое удержание влаги
Clay loam Сыпучий Clay Глина	Cyrnec Сыпучий Глинистый Глина	High-plastic Тугопластичность	—	—
Sandy loam Сыпучий Clay loam Глинистый Clay Глина	Cyrnec Сыпучий Глинистый Глина	Freeflowing / Tekuchyye	—	—
Sand Песок	—	—	—	Water not retained Влага не удерживается

- The sample of unbroken structure
Пробная неповрежденной структуры
 - The sample of broken structure
Пробная поврежденной структуры
- Уровень грунтовых вод / Ground water level
Дата замера / Date of measurement

[illegible]

Soles:
horizontal, 1:500
vertical, 1:100

Measures:
rope, 1:500
sqr., 1:100

[illegible]

- Reproduced from Client
- Not to scale

Fugro Report No. C120445

EGE NO	Soil Desctiption	Penetration		Effective Unit Weight	Design Shear Strength	Layer No.
		From	to			
		(m)	(m)	(kN/m³)	(kPa)	
1	Sandy SILT	0.0	2.6	17.7 17.7	ϕ=25°	1
2	Brown CLAY with the inclusion of carbonate	2.6	4.0	18.8 18.8	Su=145 145	2
		4.0	5.0	18.8 18.8	45 45	3
3	Firm Gray CLAY with layers of sand, plaster "nests".	5.0	7.4	9.6 9.6	35 35	4
4	Firm to stiff sandy CLAY	7.4	9.2	9.6 9.6	35 35	5
		9.2	10.8	9.6 9.6	75 75	6
3	Firm to verty stiff silty CLAY	10.8	12.4	9.6 9.6	185 185	7
		12.4	14.0	9.6 9.6	35 35	8
4	Firm sandy CLAY (Sandy SILT)	14.0	14.6	9.6 9.6	40 40	9
5a	Dense SAND	14.6	18.2	9.2 9.2	ϕ=35°	10
7	Very stiff sandy CLAY	18.2	19.8	9.3 9.3	160 160	11
6	Bluish-gray CLAY with layers of sand	19.8	21.2	9.7 9.7	180 180	12
8	Medium to Dense SAND	21.2	22.4	9.4 9.4	ϕ=30°	13
7	Stiff sandy CLAY	22.4	24.0	9.3 9.3	120 120	14
		24.0	26.0	9.3 9.3	95 95	15
6	Bluish-gray CLAY with layers of sand	26.0	28.2	9.7 9.7	120 120	16
8	Dense SAND	28.2	30.0	9.4 9.4	ϕ=35°	17
Notes:						
1) Parameters are based on API Method (2000)						
2) S _u is the design undrained shear strength of cohesive soils in kPa						
DESIGN PARAMETERS FOR AXIAL PILE CAPACITY						
CPT1247						

DESIGN PARAMETERS FOR AXIAL PILE CAPACITY

IPCI Project - Kazakhstan

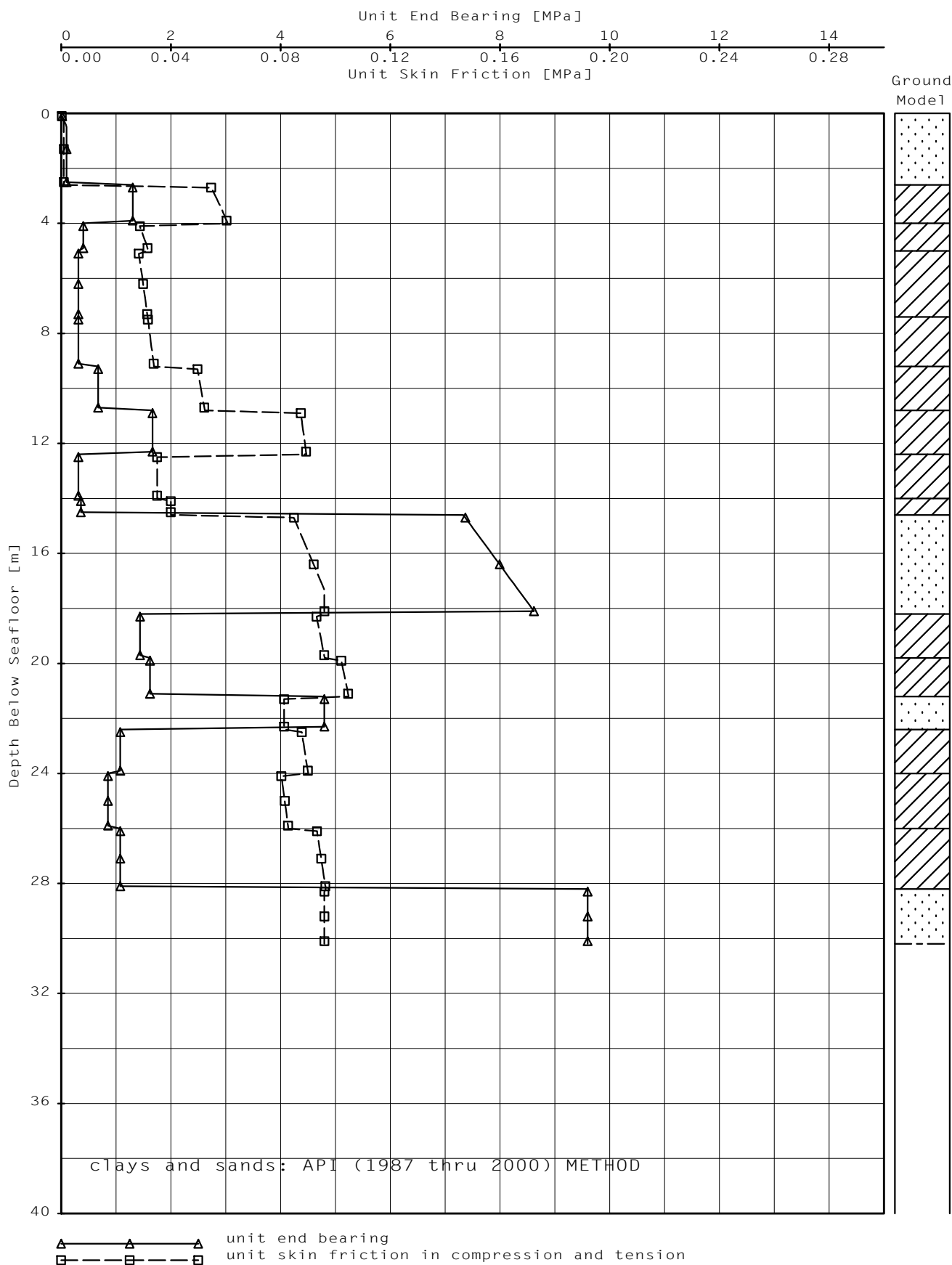
PARAMETER VALUES

dens. = v1/l/m/d/vd
= very loose/loose/medium dense/dense/very dense : sand
delta = soil-pile friction angle : sand
cu = undrained shear strength : clay
Nq = bearing capacity factor : sand
flim = limit unit skin friction : sand
qlim = limit unit end bearing pressure : sand

layer number	depth [m]	soil type	dens.	delta [deg]	cu top bot [kPa]	sub. unit wt. [kN/m3]	Nq [-]	flim [kPa]	qlim [MPa]
1	0.0	sand	1	20.0		17.7	12.0	1	0.1
2	2.6	clay			145 145	18.8			
3	4.0	clay			45 45	18.8			
4	5.0	clay			35 35	9.6			
5	7.4	clay			35 35	9.6			
6	9.2	clay			75 75	9.6			
7	10.8	clay			185 185	9.6			
8	12.4	clay			35 35	9.6			
9	14.0	clay			40 40	9.6			
10	14.6	sand	d	30.0		9.2	40.0	96	9.6
11	18.2	clay			160 160	9.3			
12	19.8	clay			180 180	9.7			
13	21.2	sand	m	25.0		9.4	20.0	81	4.8
14	22.4	clay			120 120	9.3			
15	24.0	clay			95 95	9.3			
16	26.0	clay			120 120	9.7			
17	28.2	sand	d	30.0		9.4	40.0	96	9.6
	30.2								

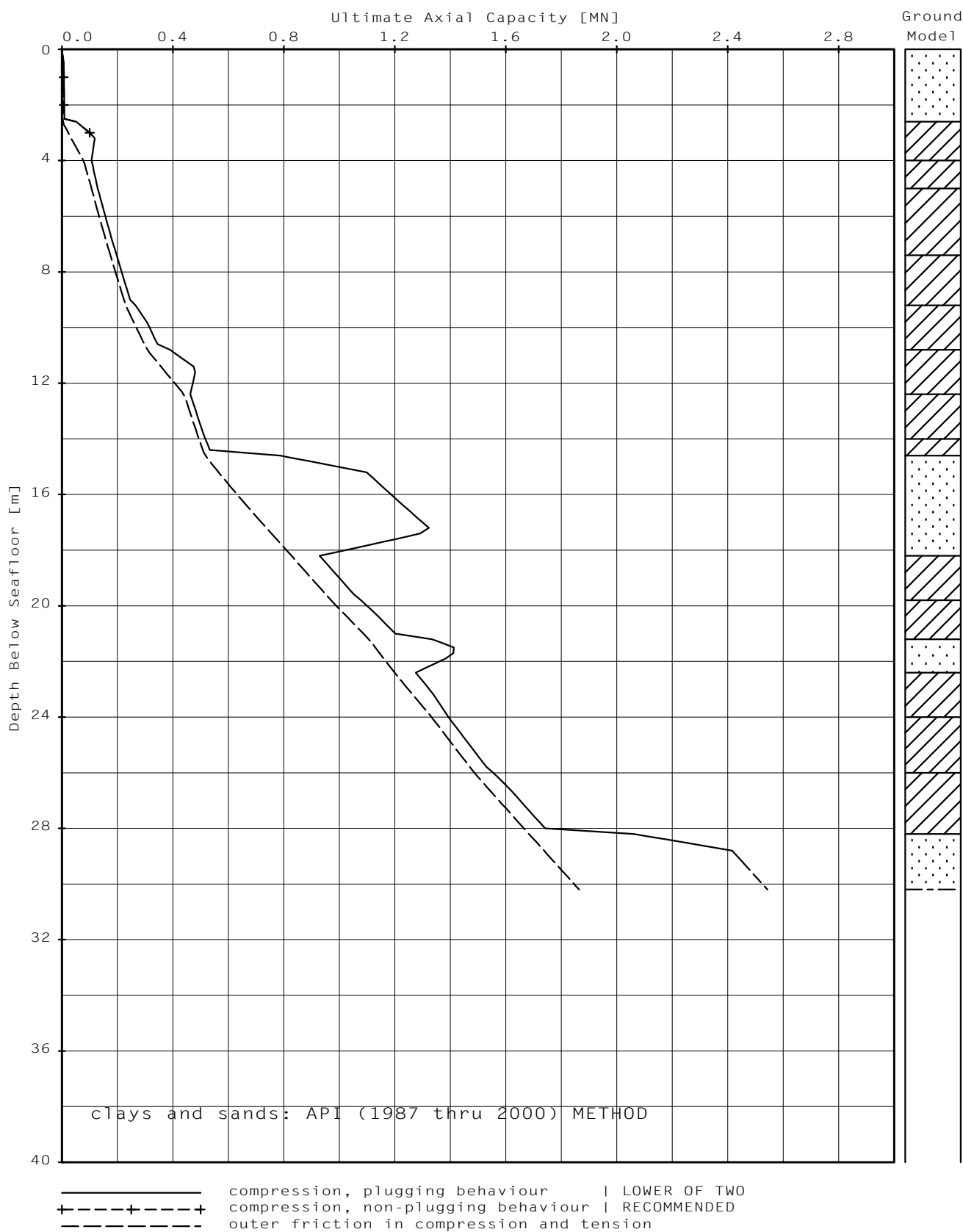
PARAMETER VALUES FOR DRIVEN PILE AXIAL CAPACITY LOCATION CPT1247

IPCI PROJECT - KAZAKHSTAN

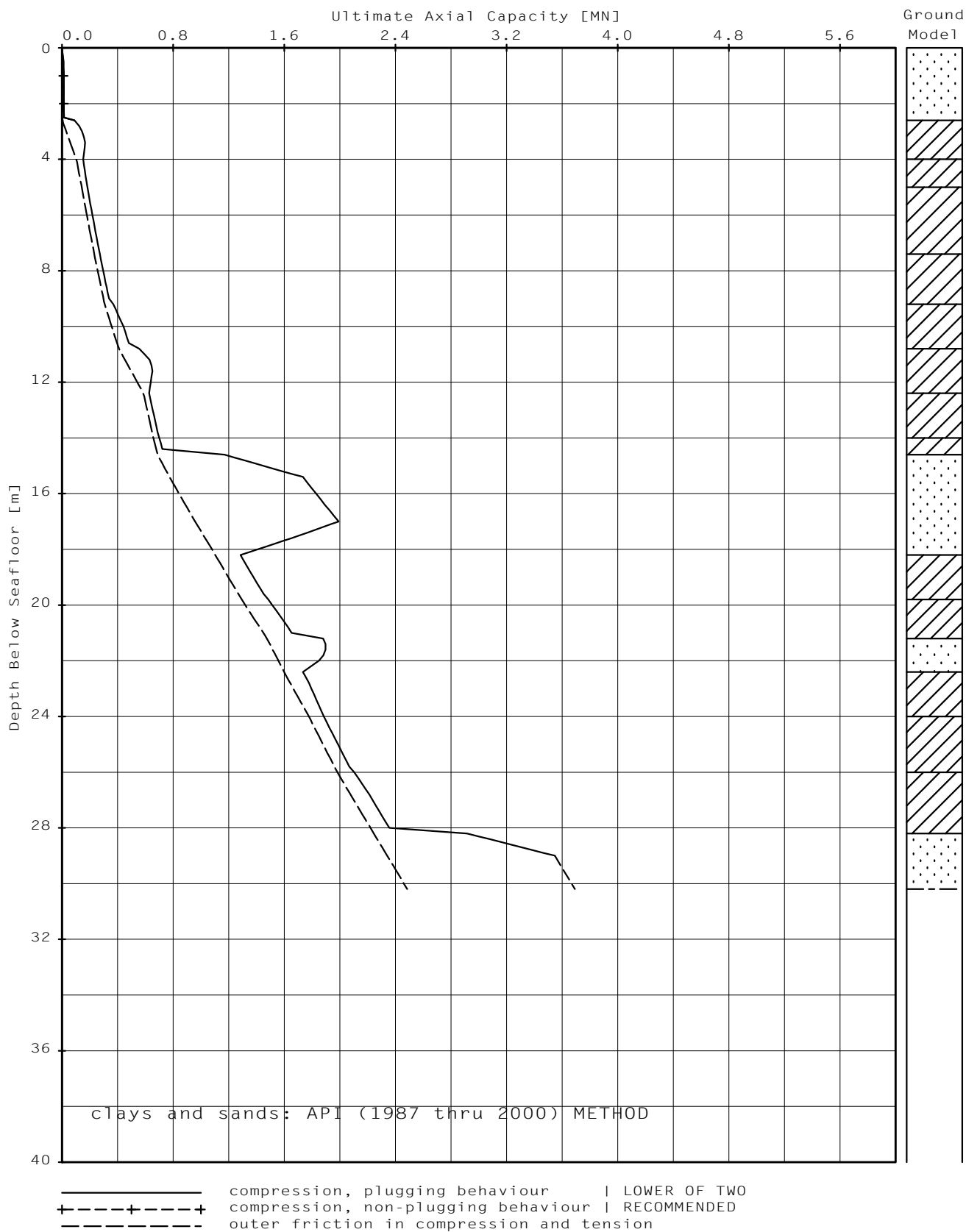


ULTIMATE UNIT END BEARING AND SKIN FRICTION FOR DRIVEN PILES
LOCATION CPT1247

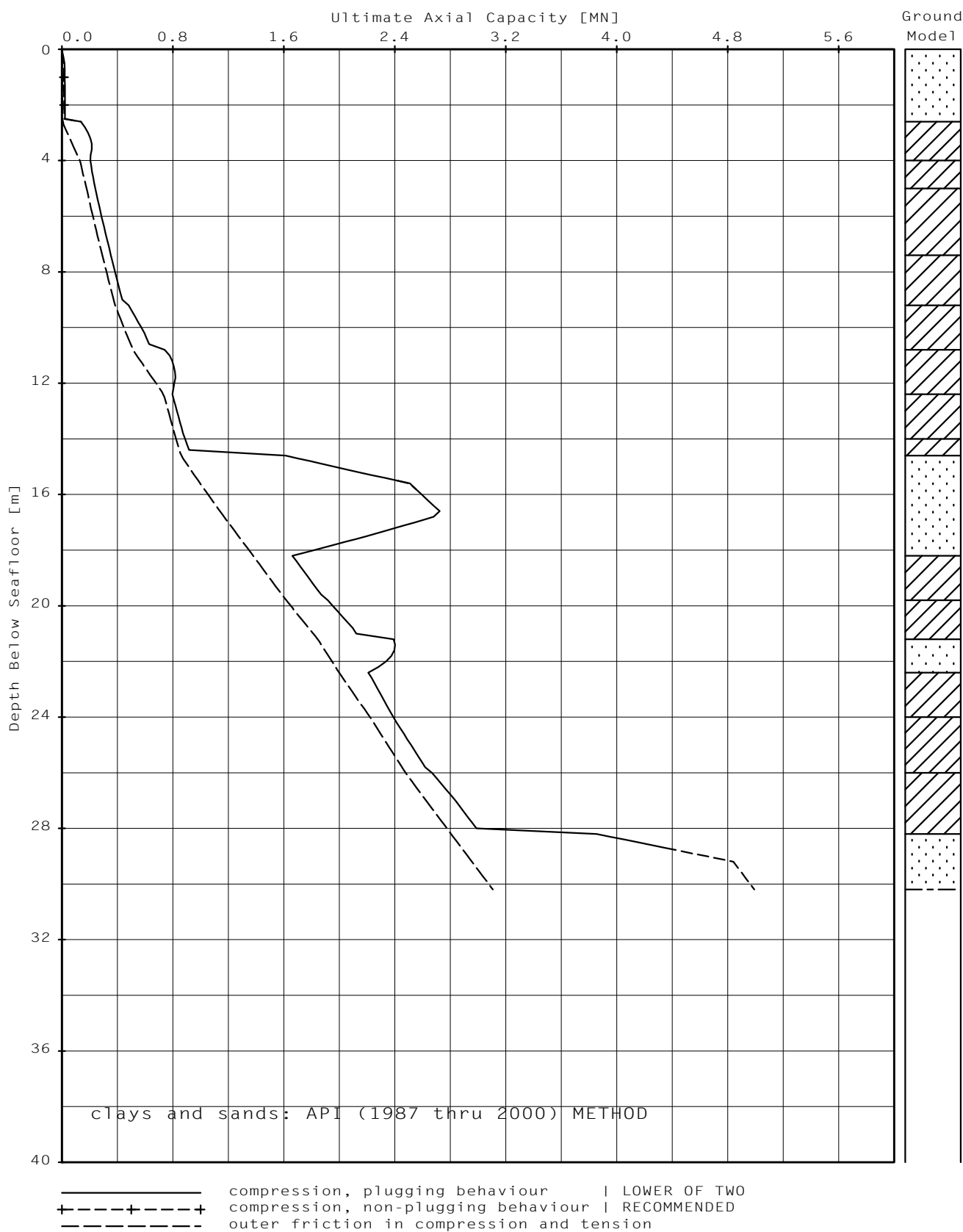
IPCI PROJECT - KAZAKHSTAN



ULTIMATE AXIAL CAPACITY FOR DRIVEN PILES
 0.3M OD X 2.75" WT PILES - LOCATION CPT1247
 IPCI PROJECT - KAZAKHSTAN



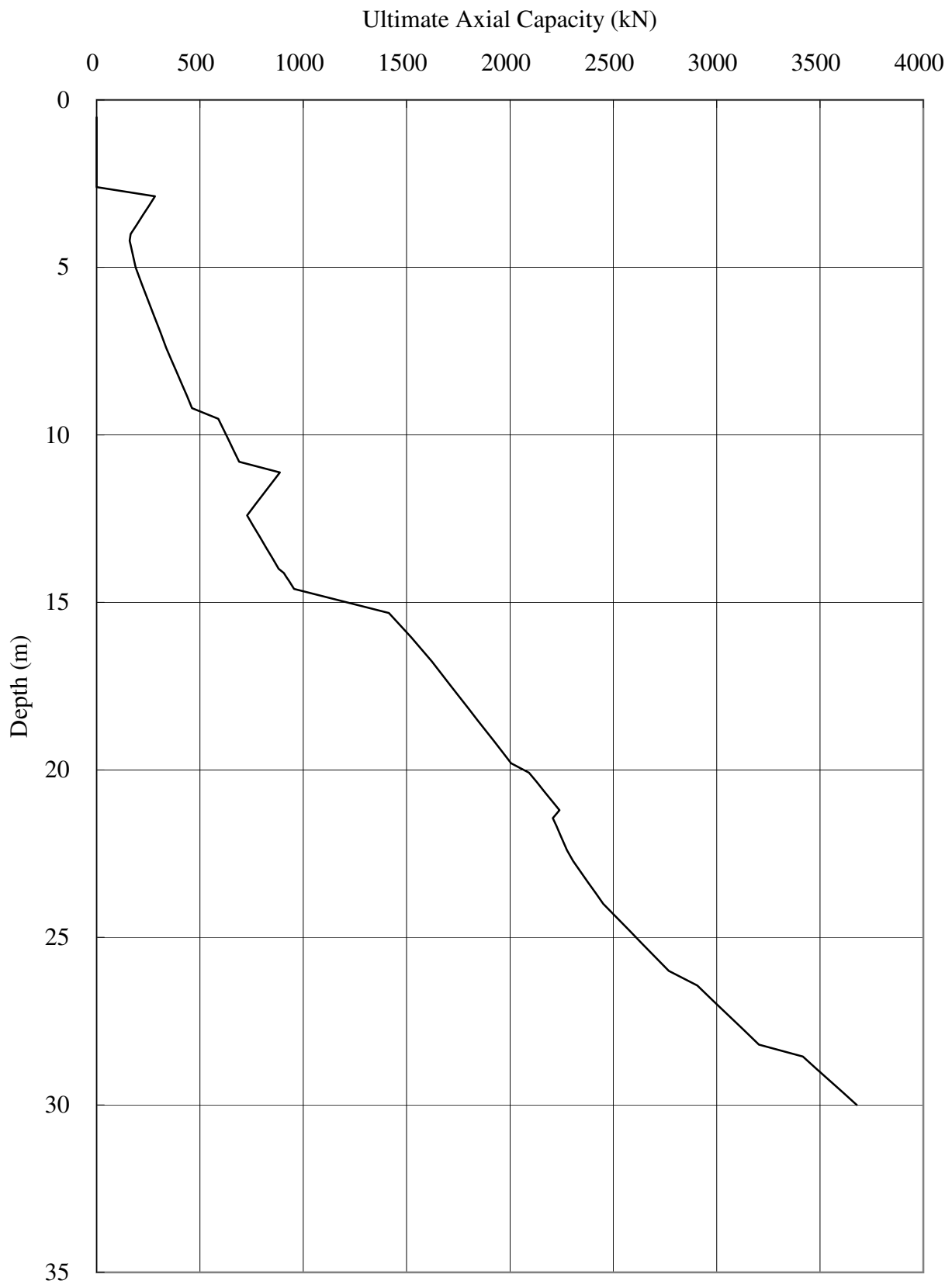
ULTIMATE AXIAL CAPACITY FOR DRIVEN PILES
 0.4M OD X 3.75" WT PILES - LOCATION CPT1247
 IPCI PROJECT - KAZAKHSTAN



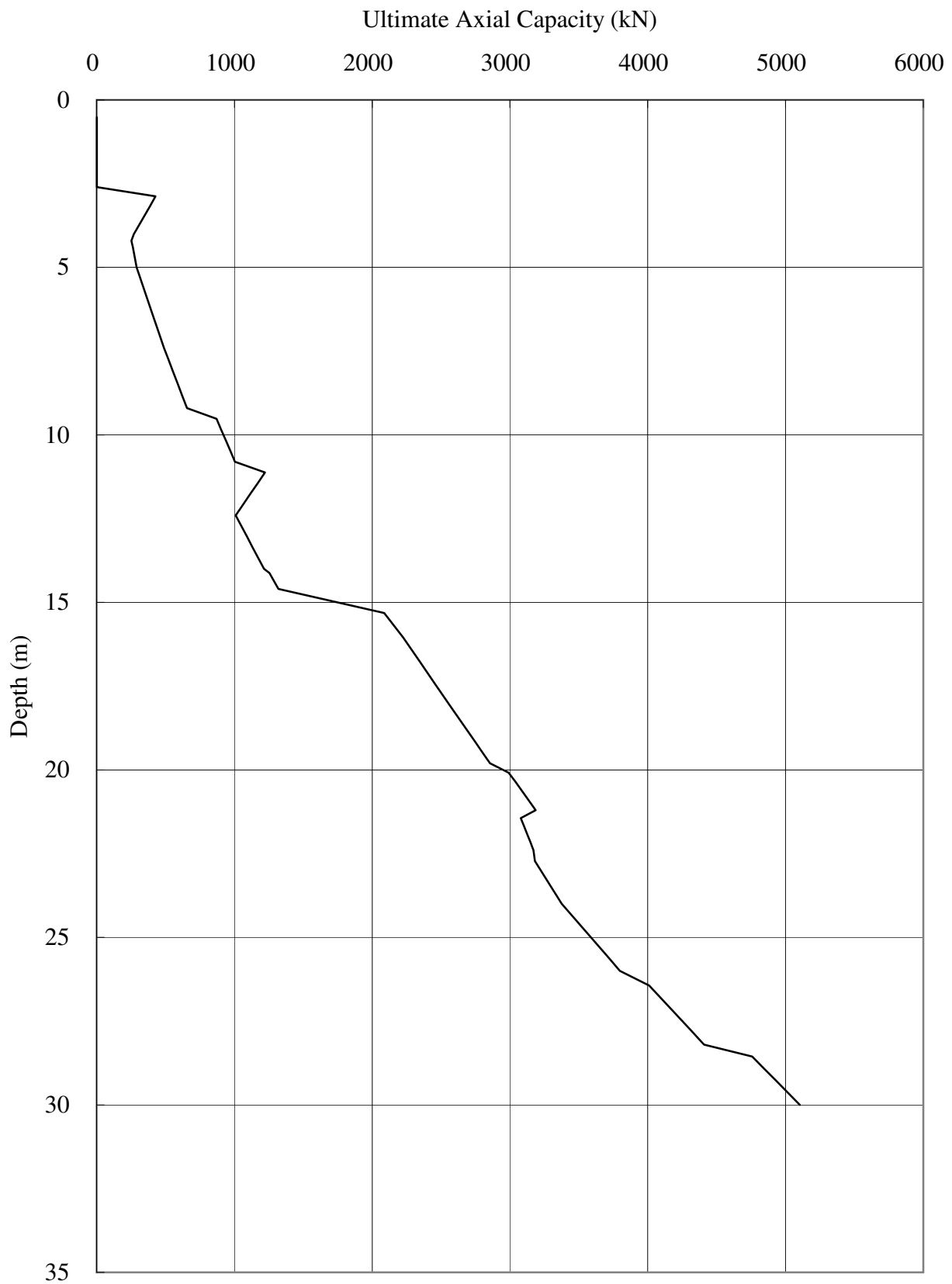
ULTIMATE AXIAL CAPACITY FOR DRIVEN PILES
 0.5M OD X 100MM WT PILES - LOCATION CPT1247
 IPCI PROJECT - KAZAKHSTAN

Layer ID	Depth(m)	Soil Type	γ (kN/m ³)	f_{max} (kPa)	f_{min} (kPa)	β	S_u (kPa)	α	Nt	q_{bmax} (kPa)
1	0.0	Sandy SILT	17.7	0.0		0.25			8	2
									8	
2	2.6	Clay	18.8	90.0		0.32	145.0		9	
							145.0		9	
2	4.0	Clay	18.8	90.0		0.32	45.0		9	
							45.0		9	
3	5.0	Clay	9.6	90.0		0.32	35.0		9	
							35.0		9	
4	7.4	F/Stiff Clay	9.6	90.0		0.30	35.0		9	
							35.0		9	
4	9.2	F/Stiff Clay	9.6	90.0		0.30	75.0		9	
							75.0		9	
3	10.8	F/Stiff Clay	9.6	90.0		0.30	185.0		9	
							185.0		9	
3	12.4	F/Stiff Clay	9.6	90.0		0.30	35.0		9	
							35.0		9	
4	14.0	F/Stiff Clay	9.6	90.0		0.30	40.0		9	
							40.0		9	
5a	14.6	Dense SAND	9.2	96.0		0.40			40	1600
									40	
7	18.2	V/stiff Clay	9.3	90.0		0.30	160.0		9	
							160.0		9	
6	19.8	V/stiff Clay	9.7	90.0		0.30	180.0		9	
							180.0		9	
8	21.2	M Dense SAND	9.4	81.3		0.40			20	1600
									20	
7	22.4	V/stiff Clay	9.3	90.0		0.30	120.0		9	
							120.0		9	
7	24.0	V/stiff Clay	9.3	90.0		0.30	95.0		9	
							95.0		9	
6	26.0	V/stiff Clay	9.7	90.0		0.30	120.0		9	
							120.0		9	
8	28.2	Dense SAND	9.4	96.0		0.40			40	1600
	30.0								40	

PARAMETER VALUES FOR BORED PILE CAPACITY
Location CPT1247
IPCI Project - Kazakhstan



ULTIMATE BEARING CAPACITY FOR BORED PILES
 600mm OD - Location CPT1247
 IPCI Project - Kazakhstan



ULTIMATE BEARING CAPACITY FOR BORED PILES
 800mm OD - Location CPT1247
 IPCI Project - Kazakhstan